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Electric Fan and Terminal Device

Abstract

An electric fan includes a motor assembly, an impeller assembly, and a diffuser assembly. The motor assembly includes an output shaft, a bracket, and a stator-rotor component. The bracket and the stator-rotor component are both sleeved on the output shaft; the bracket has an accommodation region, and the stator-rotor component is disposed in the accommodation region. The impeller assembly is disposed on a side of the stator-rotor component and is sleeved on the output shaft; the diffuser assembly is sleeved on the output shaft and is disposed between the stator-rotor component and the impeller assembly.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] The present application is a continuation application of International (PCT) Patent Application No. PCT/CN2023/108184, filed on Jul. 19, 2023, which claims priority of Chinese Patent Application No. 202211465831.7, filed on Nov. 22, 2022, the entire contents of which are hereby incorporated by reference in their entirety.

TECHNICAL FIELD

[0002] The present disclosure relates to the technical field of electric fans, and in particular to an electric fan and a terminal device.

BACKGROUND

[0003] Electric fans are widely utilized in terminal devices such as vacuum cleaners. Typically, a conventional electric fan includes an impeller assembly, a diffuser, and a motor. The diffuser is disposed downstream of the impeller assembly and is sleeved outside the motor. During operation of the electric fan, airflow enters through the inlet of the impeller assembly. Driven by the motor, the impeller assembly accelerates the airflow, which then flows out of the outlet of the impeller assembly and into the diffuser, and finally out of the outlet of the diffuser and out of the electric fan.

SUMMARY OF THE DISCLOSURE

[0004] In a first aspect, the present disclosure provides an electric fan, including a motor assembly, an impeller assembly, and a diffuser assembly; where the motor assembly includes an output shaft, a bracket, and a stator-rotor component; the bracket and the stator-rotor component are both sleeved on the output shaft; the bracket has an accommodation region, and the stator-rotor component is disposed in the accommodation region; the impeller assembly is disposed on a side of the stator-rotor component and is sleeved on the output shaft; the diffuser assembly is sleeved on the output shaft and is disposed between the stator-rotor component and the impeller assembly; wherein a part of gas passing through the diffuser assembly is configured to flow toward an outside of the bracket, and another part of the gas is configured to flow toward an interior of the stator-rotor component.

[0005] In some embodiments, a gap is defined between the stator-rotor component and the diffuser assembly in an extension direction of an axis of the output shaft.

[0006] In some embodiments, the diffuser assembly has an air outlet on an end close to the stator-rotor component; in the extension direction of the axis of the output shaft, a positive projection of the air outlet at least partially overlaps with a positive projection of the stator-rotor component.

[0007] In some embodiments, the stator-rotor component includes a rotor and a stator; the rotor is sleeved on the output shaft, and the stator is sleeved outside the rotor; in the extension direction of the axis of the output shaft, a positive projection of the air outlet at least partially overlaps with a positive projection of the stator.

[0008] In some embodiments, in the extension direction of the axis of the output shaft, the positive projection of the air outlet is a first projection, and the positive projection of the stator is a second

projection; an area where the first projection and the second projection overlap is greater than or equal to half an area of a small projection; the small projection is one of the first projection and the second projection, and the area of the small projection is less than an area of the other of the first projection and the second projection.

[0009] In some embodiments, the bracket includes a bracket body and a support member; the bracket body is sleeved on the output shaft; the support member is disposed on a side of the bracket body close to the diffuser assembly, and the support member is connected to the diffuser assembly and the bracket body, respectively.

[0010] In some embodiments, the motor assembly further includes a connecting member; the connecting member is sleeved on the output shaft and is disposed on a side of the support member away from the bracket body; the connecting member is disposed on a radial inner side of the diffuser assembly; the connecting member is connected to the support member and the diffuser assembly, respectively.

[0011] In some embodiments, the support member includes at least one support arm and a support connection; the at least one support arm is disposed on a side of the diffuser assembly close to the bracket body; the support connection is disposed on a side of the at least one support arm away from the diffuser assembly, and the support connection is connected to the at least one support arm and the bracket body, respectively.

[0012] In some embodiments, the at least one support arm is recessed toward the output shaft.

[0013] In some embodiments, the at least one support arm is a plurality of support arms, and the plurality of support arms are distributed at equal intervals along a circumferential direction of the output shaft.

[0014] In some embodiments, the motor assembly further includes a first bearing and a second bearing; the connecting member has a first bearing chamber, and the bracket body has a second bearing chamber; the first bearing and the second bearing are both sleeved on the output shaft, with the first bearing being disposed in the first bearing chamber and the second bearing being disposed in the second bearing chamber.

[0015] In some embodiments, the motor assembly further includes a plurality of elastic members; a first group of the plurality of elastic members is arranged between the first bearing and an inner wall of the first bearing chamber and respectively abuts against the first bearing and the first bearing chamber; a second group of the plurality of elastic members is arranged between the second bearing and an inner wall of the second bearing chamber and respectively abut against the second bearing and the second bearing chamber.

[0016] In some embodiments, the electric fan further includes a deflector, and a first end surface is formed on an end of the bracket adjacent to the diffuser assembly; the deflector is arranged on the first end surface; in the extension direction of the axis of the output shaft, a positive projection of the deflector at least partially overlaps with the positive projection of the air outlet.

[0017] In some embodiments, the deflector includes a deflection portion; the deflection portion protrudes toward the air outlet; in the extension direction of the axis of the output shaft, a positive projection of a point on the deflection portion furthest from the first end surface is disposed within the positive projection of the air outlet.

[0018] In some embodiments, a surface of the deflection portion close to the air outlet is curved.

[0019] In some embodiments, the electric fan further includes a mesh structure; the mesh structure covers a gap between the diffuser assembly and the stator-rotor component, and the mesh structure is connected to the diffuser assembly and the bracket, respectively.

[0020] In some embodiments, the impeller assembly includes an impeller assembly housing, a first impeller, a reflux component, and a second impeller; the first impeller, the reflux component, and the second impeller are disposed in the impeller assembly housing; the first impeller, the reflux component, and the second impeller are sequentially sleeved on the output shaft; a first flow channel is included in the first impeller, a reflux channel is formed between the reflux component

and an inner wall of the impeller assembly housing, and a second flow channel is included in the second impeller; the first flow channel, the reflux channel, and the second flow channel in a sequential communication with each other.

[0021] In some embodiments, the impeller assembly further includes an elastic sealing ring, and the elastic sealing ring includes a sealing ring body and an elastic sealing tooth; each of the sealing ring body and the elastic sealing tooth has a ring-shaped structure, and both the sealing ring body and the elastic sealing tooth are sleeved on the output shaft; the sealing ring body is disposed between the reflux component and the elastic sealing tooth and is fixedly connected to the reflux component and the elastic sealing tooth, respectively; the elastic sealing tooth is rotatably connected to the output shaft; in the extension direction of the axis of the output shaft, a width of the elastic sealing tooth is positively correlated with a distance from the elastic sealing tooth to the axis of the output shaft.

[0022] In some embodiments, the impeller assembly further includes a sealing cotton; the sealing cotton is filled between the impeller assembly housing and the first impeller and/or between the impeller assembly housing and the second impeller.

[0023] In some embodiments, the diffuser assembly includes a first axial flow diffuser and a second axial flow diffuser; the first axial flow diffuser and the second axial flow diffuser are arranged along the extension direction of the axis of the output shaft; the first axial flow diffuser is sealed and connected to the second axial flow diffuser, and the first axial flow diffuser is in communication with the second axial flow diffuser; an end of the first axial flow diffuser away from the second axial flow diffuser is sealed and connected to the impeller assembly, and the first axial flow diffuser is in communication with the impeller assembly.

[0024] In a second aspect, the present disclosure provides a terminal device, including the electric fan as described in any of the first aspects and possible embodiments thereof.

[0025] In some embodiments, the terminal device is a vacuum cleaner.

[0026] It should be understood that the above general description and the detailed description below are only illustrative and explanatory, and do not limit the present disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0027] To more clearly illustrate the technical solution in the embodiments of the present disclosure, a brief description of the drawings needed for the description of the embodiments is provided below. Obviously, the drawings described below are only some examples of the present disclosure, and for those skilled in the art, other drawings may be obtained without the expenditure of creative effort.

[0028] FIG. 1 is a structural schematic view of an electric fan according to some embodiments of the present disclosure.

[0029] FIG. 2 is a cross-sectional structural schematic view of an electric fan according to some embodiments of the present disclosure.

[0030] FIG. 3 is a structural schematic view of a bracket according to some embodiments of the present disclosure.

[0031] FIG. 4 is a schematic view of a positive projection of a partial structure of an electric fan according to some embodiments of the present disclosure.

[0032] FIG. 5 is a cross-sectional structural schematic view of an electric fan according to some embodiments of the present disclosure.

[0033] FIG. 6 is a structural schematic view of an elastic member according to some embodiments of the present disclosure.

[0034] FIG. 7 is a structural schematic view of an elastic member according to some embodiments

of the present disclosure.

[0035] FIG. **8** is a structural schematic view of a bearing chamber according to some embodiments of the present disclosure.

[0036] FIG. **9** is a cross-sectional structural schematic view of an electric fan according to some embodiments of the present disclosure.

[0037] FIG. **10** is a structural schematic view of a reflux component according to some embodiments of the present disclosure.

[0038] FIG. **11** is a structural schematic view of an elastic sealing ring according to some embodiments of the present disclosure.

[0039] FIG. **12** is a cross-sectional schematic view of FIG. **11** along line A-A.

[0040] FIG. **13** is a partial structural schematic view of an electric fan according to some embodiments of the present disclosure.

[0041] FIG. **14** is a partial structural schematic view of a diffuser assembly according to some embodiments of the present disclosure.

[0042] FIG. **15** is a cross-sectional structural schematic view of an electric fan according to some embodiments of the present disclosure.

[0043] FIG. **16** is a structural schematic view of a deflector according to some embodiments of the present disclosure.

[0044] FIG. **17** is a cross-sectional structural schematic view of an electric fan according to some embodiments of the present disclosure.

[0045] FIG. **18** is a partial structural schematic view of an electric fan according to some embodiments of the present disclosure.

[0046] FIG. **19** is a partial structural schematic view of an electric fan according to some embodiments of the present disclosure.

[0047] FIG. **20** is a cross-sectional structural schematic view of an electric fan according to some embodiments of the present disclosure.

[0048] FIG. **21** is a schematic diagram of gas flow, when there is no mesh structure, according to some embodiments of the present disclosure.

[0049] FIG. **22** is a schematic diagram of gas flow, when there is a mesh structure, according to some embodiments of the present disclosure.

[0050] FIG. **23** is a structural schematic view of an electric fan according to some embodiments of the present disclosure.

[0051] FIG. **24** is a cross-sectional structural schematic view of an electric fan according to some embodiments of the present disclosure.

[0052] FIG. **25** is a structural schematic view of a terminal device according to some embodiments of the present disclosure.

[0053] Reference numerals: **1**, motor assembly; **2**, impeller assembly; **3**, diffuser assembly; **4**, deflector; **5**, mesh structure; **11**, output shaft; **12**, bracket; **13**, stator-rotor component; **14**, connecting member; **15**, first bearing; **16**, second bearing; **17**, elastic member; **18**, electronic control board; **21**, impeller assembly housing; **22**, first impeller; **23**, reflux component; **24**, second impeller; **25**, elastic sealing ring; **26**, sealing cotton; **27**, sealing bearing; **31**, first axial flow diffuser; **32**, second axial flow diffuser; **3A**, air outlet; **41**, deflection portion; **42**, fixing portion; **43**, connecting portion; **12A**, accommodation region; **12B**, first end surface; **14A**, first bearing chamber; **14B**, first mounting groove; **121**, bracket body; **122**, support member; **131**, rotor; **132**, stator; **211**, first housing; **212**, second housing; **231**, partition plate; **232**, reflux blade; **233**, fixing protrusion; **251**, sealing ring body; **252**, elastic sealing tooth; **311**, first diffuser impeller; **312**, first diffuser housing; **321**, second diffuser impeller; **322**, second diffuser housing; **121A**, second bearing chamber; **1221**, support arm; **1222**, support connection; **1211**, body ring; **1212**, connecting arm; **231A**, mounting cavity; **251A**, positioning protrusion; **3111**, first inner shell; **3112**, first diffuser blade; **3211**, second inner shell; **3212**, second diffuser blade; **m**, axis; **O**, first direction; **01**,

electric fan; **02**, air intake apparatus; **03**, dust collection chamber; **04**, exhaust pipe.

DETAILED DESCRIPTION

[0054] Unless otherwise defined, technical or scientific terms mentioned in this text shall have the usual meaning understood by those skilled in the art. The use of the terms “first,” “second,” “third,” and the like in the description and claims of the present disclosure does not imply any order, quantity, or importance, but is merely intended to distinguish different components. Similarly, the use of terms “a” or “an”, etc. does not imply a limitation on the quantity, but rather that at least one exists. Terms such as “including” or “comprising” mean that the components or items preceding the word “including” or “comprising” cover the components or items listed after the word “including” or “comprising” and their equivalents, and do not exclude other components or items. Terms such as “connected” or “coupled” are not limited to physical or mechanical connections, but also include electrical connections, whether direct or indirect. Terms such as “above”, “below”, “left”, and “right” are intended only to indicate relative positions. When the absolute position of the described object changes, the relative position may change accordingly.

[0055] To make the purpose, technical scheme, and advantages of the present disclosure clearer, the following will further describe in detail the implementation of the present disclosure in conjunction with the accompanying drawings.

[0056] An electric fan includes an impeller assembly, a diffuser assembly, and a motor assembly. Generally, the diffuser is disposed downstream of the impeller assembly and is sleeved outside the motor. During operation of the electric fan, airflow enters through the inlet of the impeller assembly. Driven by the motor, the impeller assembly accelerates the airflow, which then flows out of the outlet of the impeller assembly and into the diffuser, and finally out of the outlet of the diffuser and out of the electric fan.

[0057] However, during the operation of the electric fan, the motor, as a driving device, will release a large amount of heat energy. Since the diffuser is sleeved outside the motor, it affects the heat dissipation of the motor, which ultimately leads to poor operational reliability and a short service life of the electric fan. The present disclosure provides an electric fan that may solve the problem of poor heat dissipation of the motor assembly in the related art. The electric fan provided by the present disclosure is described below.

[0058] FIG. 1 is a structural schematic view of an electric fan according to some embodiments of the present disclosure, and FIG. 2 is a cross-sectional structural schematic view of an electric fan according to some embodiments of the present disclosure. Referring to FIGS. 1 and 2, the electric fan includes a motor assembly 1, an impeller assembly 2, and a diffuser assembly 3, where the motor assembly 1 includes an output shaft 11, a bracket 12, and a stator-rotor component 13.

[0059] In the motor assembly 1, as shown in FIG. 2, the bracket 12 and the stator-rotor component 13 are both sleeved on the output shaft 11, and the bracket 12 and the stator-rotor component 13 are disposed on an end of the output shaft 11. The bracket 12 has an accommodation region 12A, the stator-rotor component 13 is arranged in the accommodation region 12A, and at least part of the stator-rotor component 13 is fixedly connected to the bracket 12.

[0060] As shown in FIG. 2, the impeller assembly 2 is disposed on a side of the stator-rotor component 13 and is sleeved on and connected to the output shaft 11.

[0061] As shown in FIG. 2, the diffuser assembly 3 is sleeved on the output shaft 11 and is disposed between the stator-rotor component 13 and the impeller assembly 2, and the diffuser assembly 3 is respectively connected to the bracket 12 and the impeller assembly 2. In other words, the diffuser assembly 3 is on the same side as the stator-rotor component 13.

[0062] During operation of the electric fan, the stator-rotor component 13 provides power, driving the output shaft 11 to cause the impeller assembly 2 to operate. Air is drawn in from a side of the impeller assembly 2 away from the diffuser assembly 3, is accelerated by the impeller assembly 2 and flows into the diffuser assembly 3, and flows out after being pressure-expanded by the diffuser assembly 3. The gas flowing out of the diffuser assembly 3 flows at least partially towards the

stator-rotor component **13** and eventually flows out of the electric fan.

[0063] In the solutions provided by the present embodiments, the diffuser assembly **3** is disposed between the impeller assembly **2** and the stator-rotor component **13**. During operation of the electric fan, the heat generated by the stator-rotor component **13** may be transferred to outside without being hindered by the diffuser assembly **3**, which is beneficial to improving the heat dissipation effect of the stator-rotor component **13**, thereby improving the reliability and service life of the motor assembly **1** during operation, and thus improving the reliability and service life of the entire electric fan.

[0064] The following provides a detailed description of various components of the electric fan provided in the embodiments of the present disclosure and a connection relationship between each component.

[0065] FIG. **3** is a structural schematic view of a bracket according to some embodiments of the present disclosure. As an example, with reference to FIGS. **2** and **3**, the bracket **12** includes a bracket body **121** and a support member **122**, and the bracket body **121** and the support member **122** form (or enclose to define) the accommodation region **12A**.

[0066] The bracket body **121** is sleeved on the output shaft **11** and is rotatably connected to the output shaft **11**, where the bracket body **121** may rotatably connected to the output shaft **11** by means of a bearing connection.

[0067] The support member **122** is disposed between the diffuser assembly **3** and the bracket body **121**, e.g., the support member **122** is disposed on a side of the bracket body **121** facing the diffuser assembly **3**, and the support member **122** is connected to the diffuser assembly **3** and the bracket body **121**, respectively. The support member **122** and the bracket body **121** may be integrally formed, or they may be connected by welding, screwing, or bolting. The support member **122** and the diffuser assembly **3** may be connected to each other by means of a snap fit, welding, or bolting connection.

[0068] As an example, the support member **122** may be detachably connected to the bracket body **121**, and the support member **122** may be detachably connected to the diffuser assembly **3**, which may facilitate assembly and facilitate replacement and maintenance of components.

[0069] In some embodiments, as shown in FIGS. **2** and **3**, the bracket body **121** includes a body ring **1211** and at least one connecting arm **1212**. The body ring **1211** is sleeved on the output shaft **11** and is rotatably connected to the output shaft **11**, where the body ring **1211** and the output shaft **11** may be connected by a bearing connection. The at least one connecting arm **1212** is disposed on an outer wall of the body ring **1211** and is connected to the body ring **1211**, where the connecting arm **1212** and the body ring **1211** may be integrally formed, or they may be fixed by a welding, hinging, plugging, or threading connection. As an example, as shown in FIG. **4**, when a radial dimension of the body ring **1211** is significantly different from a radial dimension of a support connection **1222**, the at least one connecting arm **1212** is bent towards the support member **122** on an end away from the body ring **1211**, to ensure connection with the support member **122**.

[0070] By virtue of this solution, it is possible to ensure that a relatively large blank region is left at the bracket body **121**, e.g., a relatively large blank region is left on a side of the stator-rotor component **13** away from the diffuser assembly **3**, such that gas may pass through the stator-rotor component **13** smoothly and dissipate heat from the stator-rotor component **13**.

[0071] In some embodiments, in an extension direction of an axis *m* of the output shaft **11**, a positive projection of the diffuser assembly **3** overlaps with a positive projection of the bracket **12**. In this case, the gas flowing out of the diffuser assembly **3** may be more smoothly diverted, such that a portion of the gas flows by an outer wall of the bracket **12** and another portion of the gas flows by the interior of the stator-rotor component **13**, which is conducive to improving the heat dissipation effect of the motor assembly **1**. The positive projection of the diffuser assembly **3** represents a projection of the diffuser assembly **3** on a plane perpendicular to the axis *m* of the output shaft **11** in the extension direction of the axis *m*, and the positive projection of the bracket **12**

represents a projection of the bracket **12** on a plane perpendicular to the axis m of the output shaft **11** in the extension direction of the axis m. The meaning of the term “positive projection” in the following is the same as or similar to the meaning of the positive projection mentioned herein and therefore will not be repeated.

[0072] In some embodiments, referring to FIGS. 2 and 3, the support member **122** includes at least one support arm **1221** and a support connection **1222**. Each of the at least one support arm **1221** has a strip-shaped or plate-shaped structure, and the support arm **1221** has a small width to reduce the blockage of the gas flowing out of the diffuser assembly **3**. The support connection **1222** has a ring-shaped structure, which may be a circular ring-shaped structure, i.e. a projection of the support connection **1222** on a cross-section perpendicular to the axis m of the output shaft **11** is a circular ring.

[0073] Referring to FIGS. 3 and 4, the at least one support arm **1221** is disposed on a side of the diffuser assembly **3** close to the bracket body **121**, and each support arm **1221** is respectively connected to the diffuser assembly **3**. The support connection **1222** is disposed on a side of the at least one support arm **1221** away from the diffuser assembly **3**, and the support connection **1222** is respectively connected to the at least one support arm **1221** and the bracket body **121**. In particular, the support connection **1222** and the support arm **1221** may be integrally formed, or they may be fixedly connected by welding or gluing; the support connection **1222** and the bracket body **121** may be integrally formed, or they may be connected by a welding, threaded, or bolt connection.

[0074] In some embodiments, the support member **122** may include only a single support arm **1221**, such that the obstruction to the gas may be minimized.

[0075] In other embodiments, the support member **122** may include multiple support arms **1221**, in which case the multiple support arms **1221** are distributed at equal intervals along a circumferential direction of the output shaft **11**. When the support connection **1222** has a circular ring-shaped structure, the multiple support arms **1221** may be considered to be distributed at equal intervals around the circumference of the support connection **1222**. This solution may be conducive to improving the stability and reliability of the electric fan after assembly. In some embodiments, the multiple support arms **1221** may be distributed at non-equal intervals along the circumferential direction of the output shaft **11**.

[0076] As an example, FIG. 4 is a schematic view of a positive projection of a partial structure of an electric fan according to some embodiments of the present disclosure. referring to FIG. 4, in the extension direction of the axis m of the output shaft **11**, a positive projection of the support arm **1221** overlaps with a positive projection of the connecting arm **1212**. Specifically, a radial dimension of the support connection **1222** of the support member **122** is similar to a radial dimension of the bracket body **121**, the number of support arms **1221** of the support member **122** is the same as the number of connecting arms **1212** of the bracket body **121**, and the positive projection of the support arm **1221** overlaps with the positive projection of the connecting arm **1212** in the extension direction of the axis m of the output shaft **11**.

[0077] This solution is advantageous in that it facilitates ensuring the coaxiality between the support member **122** and the bracket body **121** during assembly, thereby ensuring the reliability of the electric fan during operation; further, it is advantageous in that it reduces the overall blockage of the bracket **12** to the gas, minimizes the aerodynamic drag of the bracket **12**, and allows more gas to pass through the interior of the stator-rotor component **13**, thereby further enhancing the heat dissipation effect.

[0078] In some embodiments, the support member **122** may have a mesh structure or a shell structure. When the support member **122** has a mesh structure or a hollow shell structure (a shell with through holes), a portion of the gas blown out from the diffuser assembly **3** flows into the interior of the stator-rotor component **13**, and the other portion of the gas may flow out of the support member **122** through the pores in the mesh structure. When the support member **122** is a sealed shell structure (e.g., a shell without through holes), all the gas flowing out of the diffuser

assembly 3 flows to the stator-rotor component 13. No limitations are placed on the specific structure of the support member 122. The shape of the connecting arm 1212 is only required to ensure that the gas passing through the stator-rotor component 13 can flow outwards from the connecting arm 1212, such as a mesh structure or a hollow shell structure, without any limitations. [0079] Referring to FIG. 2, a gap is defined between the stator-rotor component 13 and the diffuser assembly 3 in the extension direction of the axis m of the output shaft 11 (e.g., the axial direction of the output shaft 11). The gas flowing out of the diffuser assembly 3 diffuses in this gap, where a portion of the gas flows past the outer wall of the stator-rotor component 13, taking away the heat from the outer wall, and a portion of the gas flows past the interior of the stator-rotor component 13, taking away the heat from the interior, which is more conducive to improving the heat dissipation effect of the stator-rotor component 13, thereby further improving the operational reliability and service life of the entire electric fan. Moreover, the gap may slow down the gas flowing out of the diffuser assembly 3 to a certain extent, thereby reducing the noise generated by the gas impacting the stator-rotor component 13.

[0080] Referring to FIG. 2, the stator-rotor component 13 includes a rotor 131 and a stator 132. The rotor 131 is fixed to the output shaft 11 by means of an interference fit or a key connection, and the stator 132 is fixedly connected to an inner wall of the bracket 12. As an example, as shown in FIG. 2, the stator 132 is sleeved outside the rotor 131, and an air gap is defined between the stator 132 and the rotor 131. In this case, an inner wall of the stator 132 is opposite to an outer wall of the rotor 131. In the extension direction of the axis m of the output shaft 11, a positive projection of an air outlet 3A at least partially overlaps with a positive projection of the stator 132. This solution may allow the dimensions of the motor assembly 1 in the extension direction of the axis m of the output shaft 11, i.e. the axial dimensions of the motor assembly, to be controlled, thereby reducing the axial length of the electric fan.

[0081] In some embodiments, the stator 132 and the rotor 131 are both sleeved on the output shaft 11, and a gap is defined between the stator 132 and the output shaft 11. In the extension direction of the axis m of the output shaft 11, an air gap is defined between the stator 132 and the rotor 131, and an end surface of the stator 132 is opposite an end surface of the rotor 131. This solution is advantageous for reducing the radial dimensions of the electric fan.

[0082] In some embodiments, the diffuser assembly 3 has an air outlet 3A on an end close to the stator-rotor component 13, and the positive projection of the air outlet 3A in the extension direction of the axis m of the output shaft 11 at least partially overlaps with the positive projection of the stator-rotor component 13.

[0083] As an example, referring to FIG. 2, when the stator 132 is sleeved outside the rotor 131, the air outlet 3A is axially aligned with the stator 132. Specifically, in the extension direction of the axis m of the output shaft 11, the positive projection of the air outlet 3A is a first projection, and the positive projection of the stator 132 is a second projection. The area where the first projection and the second projection overlap is greater than or equal to half the area of the projection with a smaller area among the first projection and the second projection. For example, the area of the first projection is 10 cm^2 , and the area of the second projection is 15 cm^2 , it is necessary to ensure that the area where the first projection and the second projection overlap is greater than or equal to 5 cm^2 . This solution is conducive to ensuring that the gas flowing out of the air outlet 3A of the diffuser assembly 3 diffuses into the interior of the stator-rotor component 13, thereby improving the heat dissipation effect on the stator-rotor component 13.

[0084] In some embodiments, a radial dimension of the diffuser assembly 3 is approximately equal to a radial dimension of the motor assembly 1. In this case, in the extension direction of the axis m of the output shaft 11, an outer contour of the positive projection of the air outlet 3A may coincide with an outer contour of the positive projection of the stator 132.

[0085] In some embodiments, referring to FIGS. 2 and 3, the motor assembly 1 further includes a connecting member 14. The connecting member 14 is sleeved on the output shaft 11 and is

disposed on a side of the support arm **1221** away from the support connection **1222**. The diffuser assembly **3** is sleeved on the connecting member **14** and is connected to the connecting member **14**. In this case, the support arm **1221** is recessed toward the output shaft **11**, and an end of the support arm **1221** away from the support connection **1222** is connected to the connecting member **14**. This solution may reduce the occupation of the flow channel of the diffuser assembly **3**, thereby reducing the blocking effect on the gas and improving the gas flow efficiency. Furthermore, the support arm **1221** has a curved shape, which may divert part of the gas flow to the outside of the bracket body **121**, thereby improving the heat dissipation effect of the stator-rotor component. [0086] In some embodiments, when the motor assembly **1** further includes the connecting member **14**, the end of the support arm **1221** away from the support connection **1222** may be directly connected to the diffuser assembly **3**. The connection relationship between the support arm **1221** and the diffuser assembly **3** is not limited in any way.

[0087] As an example, as shown in FIG. **4**, when the support arm **1221** is recessed toward the output shaft **11** and connected to the connecting member **14**, in the extension direction of the axis **m** of the output shaft **11**, the positive projection of the support arm **1221** overlaps with the positive projection of the connecting arm **1212**, thereby minimizing the aerodynamic drag of the bracket **12** and allowing more gas to pass through the interior of the stator-rotor component **13**, and thus further improving the heat dissipation effect.

[0088] FIG. **5** is a cross-sectional structural schematic view of an electric fan according to some embodiments of the present disclosure. In some embodiments, referring to FIGS. **3** and **5**, the motor assembly **1** further includes a first bearing **15** and at least one elastic member **17**, and the connecting member **14** has a first bearing chamber **14A**. The output shaft **11** is disposed in the first bearing chamber **14A**, and a gap is defined between the output shaft **11** and an inner wall of the first bearing chamber **14A**. The first bearing **15** is sleeved on the output shaft **11** and is disposed in the first bearing chamber **14A**, where the first bearing **15** and the output shaft **11** may be connected by a press fit or a key connection, and a gap is defined between the first bearing **15** and the inner wall of the first bearing chamber **14A**. The at least one elastic member **17** is disposed between the first bearing **15** and the inner wall of the first bearing chamber **14A**, and abuts against the first bearing **15** and the first bearing chamber **14A** respectively to provide a preload force required for assembly of the first bearing **15**. A gap may be defined between an outer wall of the first bearing **15** and the inner wall of the first bearing chamber **14A**, or the outer wall of the first bearing **15** is in contact with the inner wall of the first bearing chamber **14A** without generating an interaction force. As an example, the total preload force (i.e. total spring force, which refers to the sum of the spring forces provided by all elastic members **17** in a certain direction) that can be provided by the at least one elastic member **17** is in a range of 10 N to 20 N. The total spring force of the at least one elastic member **17** may be determined according to the preload force required for the bearing and output shaft to operate normally in actual products, which is not limited herein.

[0089] In the relevant art, the output shaft is fixedly connected to the bearing by means of a press fit or key connection, while the bearing is connected to the bearing chamber by a clearance fit. When the speed of the output shaft increases or when the length of the output shaft increases, the output shaft is prone to stress imbalance, excessive vibration, and serious deformation during high-speed rotation, which ultimately leads to product failure.

[0090] In the solutions provided in the present disclosure, the at least one elastic member **17** is arranged between the first bearing **15** and the first bearing chamber **14A**, and the at least one elastic member **17** respectively abuts against the first bearing **15** and the inner wall of the first bearing chamber **14A**, thereby providing a certain preload force for the first bearing **15**. The preload force is conducive to ensuring the coaxiality between the first bearing **15** and relevant components during assembly. Furthermore, during the high-speed rotation of the output shaft **11**, the at least one elastic member **17** may generate elastic force through deformation to reduce the deflection and vibration of the output shaft **11**, thereby preventing stress imbalance, improving the vibration-reducing

effect, which may thus help to prevent the output shaft **11** from failing and increase the service life of the product.

[0091] FIG. **6** is a structural schematic view of an elastic member according to some embodiments of the present disclosure, and FIG. **7** is a structural schematic view of an elastic member according to some embodiments of the present disclosure. Referring to FIGS. **6** and **7**, each elastic member **17** may have a ring-shaped structure. The elastic member **17** may be an O-shaped ring as shown in FIG. **6**, or a square ring (with an inner circular surface and an outer square surface) as shown in FIG. **7**. The elastic member **17** is sleeved outside the first bearing **15**, and a surface of the elastic member **17** away from the first bearing **15** abuts against the inner wall of the first bearing chamber **14A**. There are no limitations on the shape of the elastic member **17**, as long as the elastic member **17** is sleeved outside the first bearing **15**.

[0092] In some embodiments, the elastic member **17** may have a block structure, in which case the motor assembly **1** includes multiple elastic members **17**, such as 3, 4, 8, etc. in number. The multiple elastic members **17** are distributed equally in a circumferential direction on the outer wall of the first bearing **15**, and each elastic member **17** respectively abuts against the first bearing **15** and the first bearing chamber **14A** to provide a corresponding preload force.

[0093] FIG. **8** is a structural schematic view of a bearing chamber according to some embodiments of the present disclosure. In some embodiments, as shown in FIG. **8**, the inner wall of the first bearing chamber **14A** defines at least one first mounting groove **14B**, and the number of the at least one first mounting groove **14B** is equal to the number of the at least one elastic member **17**. The at least one elastic member **17** is matched with the at least one first mounting groove **14B**, and a portion of each elastic member **17** is disposed in a corresponding first mounting groove **14B**. During assembly, the first bearing **15** squeezes the elastic member **17** to deform it, and a gap is defined between the outer wall of the first bearing **15** and the inner wall of the first bearing chamber **14A**, or the outer wall of the first bearing **15** is in contact with the inner wall of the first bearing chamber **14A** generating an interaction force (e.g., without friction).

[0094] As an example, when the elastic member **17** is an O-shaped ring, the first mounting groove **14B** is an O-shaped ring groove, and each O-shaped ring is partially snapped in the first mounting groove **14B**. Specifically, an outer wall arc of the O-shaped ring (e.g., the elastic member **17**) fits a bottom arc of the O-shaped ring groove (e.g., the first mounting groove **14B**), two end surfaces of the O-shaped ring fit two groove walls of the O-shaped ring groove, and an inner wall of the O-shaped ring protrudes from the first bearing chamber **14A** toward the output shaft **11**, e.g., a radial thickness of the O-shaped ring is greater than a depth of the O-shaped ring groove.

[0095] In some embodiments, when the elastic member **17** is a square ring as shown in FIG. **7**, the first mounting groove **14B** is a square ring groove matching the square ring. In this case, the assembly relationship between the square ring and the square ring groove is similar to the assembly relationship between the O-shaped ring and the O-shaped ring groove described above, which will not be described in detail here.

[0096] In some embodiments, when the elastic member **17** has a block structure, the first mounting groove **14B** is a groove matching the block structure (e.g., the elastic member **17**). In this case, the assembly relationship between the block structure and the groove is similar to the assembly relationship between the O-shaped ring and the O-shaped ring groove described above, which will not be described in detail here.

[0097] There are no limitations on the shape of the first bearing chamber **14A**, as long as the first bearing chamber **14A** matches the elastic member **17** and the surface the elastic member **17** away from the bottom of the first mounting groove **14B** protrudes from the first bearing chamber **14A**.

[0098] By virtue of this solution, at least part of the elastic member **17** is disposed in the first mounting groove **14B**, which is advantageous for preventing the elastic member **17** from being displaced along the extension direction of the axis *m* of the output shaft **11** during high-speed rotation of the output shaft **11**. Since the elastic member **17** abuts against the outer wall of the first

bearing **15**, a certain frictional force may be generated, such that, during high-speed rotation of the output shaft **11**, it is advantageous to prevent the first bearing **15** from displacing along the extension direction of the axis **m** of the output shaft **11**, thereby ensuring the working stability of the motor assembly **1** and the electric fan.

[0099] In some embodiments, the outer wall of the first bearing **15** defines at least one first bearing mounting groove, the number of the at least one first bearing mounting groove being equal to the number of the at least one elastic member **17**. The at least one elastic member **17** is matched to the at least one first bearing mounting groove, and a portion of each elastic member **17** is disposed in a corresponding first bearing mounting groove.

[0100] For example, when the elastic member **17** is an O-shaped ring, the first bearing mounting groove is an O-shaped ring groove, and a portion of each O-shaped ring is snapped in the first bearing mounting groove. Specifically, an inner wall arc of the O-shaped ring (e.g., elastic member **17**) fits a bottom arc of the O-shaped ring groove (e.g., first bearing mounting groove), two end surfaces of the O-shaped ring fit two groove walls of the O-shaped ring groove, and an outer wall of the O-shaped ring protrudes from the first bearing **15** in a direction away from output shaft **11**, e.g., a radial thickness of the O-shaped ring is greater than a depth of the O-shaped ring groove. After assembly, a gap is defined between the outer wall of the first bearing **15** and the inner wall of the first bearing chamber **14A**, or the outer wall of the first bearing **15** is in contact with the inner wall of the first bearing chamber **14A** without generating an interaction force (e.g., without friction).

[0101] In cases where the elastic member **17** is a square ring or the elastic member **17** has a block structure, the assembly relationship between the elastic member **17** and the first bearing mounting groove is similar to the assembly relationship when the elastic member **17** is an O-shaped ring, which will not be described in detail here.

[0102] In some embodiments, taking the elastic member **17** as an O-shaped ring as an example, when a ratio of an axial length of the first bearing **15** to an outer diameter of the first bearing **15** is less than 0.5, the motor assembly **1** may include only one elastic member **17**. The axial length of the first bearing **15** refers to a length of the first bearing **15** in the extension direction of the axis **m** of the output shaft **11**, and may also be understood as a width of the first bearing **15**. For example, when the axial length of the first bearing **15** is 3 mm and the outer diameter of the first bearing **15** is 12 mm, the motor assembly **1** may include only one O-shaped ring. For another example, when the axial length of the first bearing **15** is 6 mm and the outer diameter of the first bearing **15** is 12 mm, the motor assembly **1** may include two O-shaped rings.

[0103] As an example, the axial length of the first bearing **15** ranges from 3 mm to 6 mm, and the outer diameter of the first bearing **15** ranges from 12 mm to 16 mm. The dimensions of the first bearing **15** may be set according to the actual product requirements, which is not limited herein.

[0104] This solution is advantageous in that it may reasonably control the number of elastic members **17**, preventing the occurrence of a situation in which the vibration-damping effect is reduced due to an excessive number of elastic members **17**, and is further advantageous in that it may control costs and reduce assembly complexity. Herein, only an exemplary method for determining the number of elastic members **17** is provided, and no limitations are imposed on the method for determining the number of elastic members **17**.

[0105] As an example, the elastic member **17** is made of a soft rubber material, such as nitrile rubber, silicone, polyethylene or polypropylene.

[0106] As an example, an axial length of the elastic member **17**, that is, a thickness of the elastic member **17**, ranges from 0.1 mm to 0.15 mm. The dimensions of the elastic member **17** may be set according to the actual product requirements, which is not limited herein.

[0107] As shown in FIGS. **3** and **5**, the motor assembly **1** may further include a second bearing **16** and multiple elastic members **17**, and the bracket **12** has a second bearing chamber **121A**.

Specifically, the body ring **1211** of the bracket body **121** of the bracket **12** has the second bearing

chamber **121A**. The output shaft **11** is disposed in the second bearing chamber **121A**, and a gap is defined between an outer wall of the output shaft **11** and an inner wall or a surface of the second bearing chamber **121A**. The second bearing **16** is sleeved on the output shaft **11** and is disposed in the second bearing chamber **121A**. Some of the multiple elastic members **17** are arranged between the second bearing **16** and an inner wall of the second bearing chamber **121A**, and these elastic members **17** respectively abut against the second bearing **16** and the second bearing chamber **121A** to provide a preload force required for the assembly of the second bearing **16**. A gap is defined between an outer wall of the second bearing **16** and the inner wall of the second bearing chamber **121A**, or the outer wall of the second bearing **16** is in contact with the inner wall of the second bearing chamber **121A** without generating an interaction force.

[0108] As an example, when each elastic member **17** is an O-shaped ring and the motor assembly **1** includes three O-shaped rings, two O-shaped rings are sleeved on the first bearing **15**, the other O-shaped ring is sleeved on the second bearing **16**; or, one O-shaped ring is sleeved on the first bearing **15**, and the other two O-shaped rings are sleeved on the second bearing **16**.

[0109] As an example, when each elastic member **17** has a block structure and the motor assembly **1** includes ten elastic members **17**, five elastic members **17** are uniformly distributed along a circumferential direction of the outer wall of the first bearing **15**, five elastic members **17** are uniformly distributed along a circumferential direction of the outer wall of the second bearing **16**; or, four elastic members **17** are uniformly distributed along the circumferential direction of the outer wall of the first bearing **15**, and six elastic members **17** are uniformly distributed along the circumferential direction of the outer wall of the second bearing **16**.

[0110] In some embodiments, when a dimension of the first bearing **15** is the same as a dimension of the second bearing **16**, a dimension of the elastic member **17** disposed on the outer wall of the first bearing **15** is the same as a dimension of the elastic member **17** disposed on the outer wall of the second bearing **16**. In other embodiments, when the dimension of the second bearing **16** is different from the dimension of the second bearing **16**, the dimension of the elastic member **17** disposed on the outer wall of the first bearing **15** is different from the dimension of the elastic member **17** disposed on the outer wall of the second bearing **16**.

[0111] In some embodiments, referring to FIG. **8**, the inner wall of the second bearing chamber **121A** defines at least one second mounting groove **121B**, the number of second mounting grooves **121B** being equal to the number of the elastic members **17** arranged on the outer wall of the second bearing **16**, and the second mounting grooves **121B** being matched with the elastic members **17**. The form structure of the second mounting groove **121B** is the same as or similar to the form structure of the first mounting groove **14B**, the assembly relationship between the second mounting groove **121B** and the elastic member **17** is the same as or similar to the assembly relationship between the first mounting groove **14B** and the elastic member **17**, which will not be described in detail here.

[0112] In some embodiments, the outer wall of the second bearing **16** defines at least one second bearing mounting groove, the number of second bearing mounting grooves being equal to the number of the elastic members **17** arranged on the outer wall of the second bearing **16**, and the second bearing mounting grooves being matched with the elastic members **17**. The form structure of the second bearing mounting groove is the same as or similar to that of the first bearing mounting groove, and the assembly relationship between the second bearing mounting groove and the elastic member **17** is the same as or similar to that between the first bearing mounting groove and the elastic member **17**, which will not be described in detail here.

[0113] FIG. **9** is a cross-sectional structural schematic view of an electric fan according to some embodiments of the present disclosure. As shown in FIG. **9**, the impeller assembly **2** may include an impeller assembly housing **21**, a first impeller **22**, a reflux component **23**, and a second impeller **24**. The impeller assembly housing **21** is opened on both ends along the direction of the axis **m**, where the opening of the impeller assembly housing **21** away from the diffuser assembly **3** serves

as an air inlet of the electric fan, and the opening of the impeller assembly housing **21** close to the diffuser assembly **3** is connected to the diffuser assembly **3**. The impeller assembly housing **21** is sealed and connected to the diffuser assembly **3** to ensure that the air flow accelerated by the impeller assembly **2** flows entirely into the diffuser assembly **3**.

[0114] As an example, as shown in FIG. **9**, the first impeller **22**, the reflux component **23** and the second impeller **24** are disposed in the impeller assembly housing **21**. The first impeller **22**, the reflux component **23**, and the second impeller **24** are sequentially sleeved on the output shaft **11**. The first impeller **22** or the second impeller **24** is fixedly connected to the output shaft **11**, and the reflux component **23** is rotatably connected to the output shaft **11**. Sealing cotton is filled between the first impeller **22** and the impeller assembly housing **21**, and filled between the second impeller **24** and the impeller assembly housing **21**. The sealing cotton may prevent gas from flowing out of a gap between the first impeller **22** and the impeller assembly housing **21** and a gap between the second impeller **24** and the impeller assembly housing **21**, which is conducive to improving the efficiency of gas flow and thus the efficiency of the electric fan. There are no limitations on the connection of each of the first impeller **22**, the reflux component **23**, and the second impeller **24**, with the output shaft **11**.

[0115] As an example, as shown in FIG. **9**, the first impeller **22** has a first flow channel, and the reflux component **23** and an inner wall of the impeller assembly housing **21** form a reflux channel. The second impeller **24** has a second flow channel, and the first flow channel, the reflux channel, and the second flow channel in a sequential communication. The gas enters the first flow channel through the air inlet of the electric fan (e.g., the opening of the impeller assembly housing **21** away from the diffuser assembly **3**), is accelerated after passing through the first impeller **22**, enters the reflux channel, enters the second flow channel after refluxing or diverting through the reflux component, is accelerated after passing through the second impeller, and enters the diffuser assembly **3**.

[0116] In some embodiments, the impeller assembly **2** may include only one impeller or may include more than two impellers. In both cases, the impeller assembly **2** is similar to the above case with the first impeller **22** and the second impeller **24**, which will not be described in detail herein.

[0117] In the related art, the diffuser assembly is sleeved on the motor assembly. In this case, the outer diameter of the electric fan is equivalent to the sum of the outer diameter of the motor assembly and the radial dimension of the diffuser assembly. However, in the embodiments of the present disclosure, which provides a multi-stage impeller, the radial dimension of the electric fan is equivalent to the radial dimension of the motor assembly or the radial dimension of the diffuser assembly. The electric fan in the proposed solutions has a smaller radial dimension compared to the electric fan in the related art, thereby facilitating product miniaturization. Moreover, the solution with the multi-stage impeller is conducive to improving the suction power of the electric fan to meet product development needs, and is conducive to reducing the radial dimension of the electric fan under the same suction power, thereby improving the applicability of the electric fan.

[0118] In some embodiments, as shown in FIG. **9**, the impeller assembly housing **21** includes a first housing **211** and a second housing **212**, the second housing **212** being disposed between the first housing **211** and the diffuser assembly **3** and being sealed and connected to the first housing **211** and the diffuser assembly **3**, respectively. The second housing **212** and the first housing **211** may be detachably connected by welding, snap-fitting, etc. The first impeller **22** is disposed in the first housing **211**, and the reflux component **23** and the second impeller **24** are disposed in the second housing **212**. The specific positional relationship is similar to that described above, which will not be described in detail herein. The solution with a detachable connection between the first housing **211** and the second housing **212** is advantageous in that it may reduce the difficulty of assembling the impeller assembly **2**, thereby improving production efficiency and facilitating subsequent maintenance.

[0119] In some embodiments, the first impeller **22** and the reflux component **23** are disposed in the

first housing **211**, and the second impeller **24** is disposed in the second housing **212**. For this structure, it will not be repeated herein.

[0120] FIG. **10** is a structural schematic view of a reflux component according to some embodiments of the present disclosure. Referring to FIGS. **9** and **10**, the reflux component **23** includes a partition plate **231** and multiple reflux blades **232**. The partition plate **231** is sleeved on the output shaft **11**. The multiple reflux blades **232** are arranged on a side of the partition plate **231** close to the diffuser assembly **3**, are unevenly distributed along the circumferential direction of the output shaft **11**, and are connected to the partition plate **231**. The reflux blades **232** and the partition plate **231** may be integrally formed, or they may be connected to each other by welding, gluing, or plugging, etc.

[0121] In some embodiments, as shown in FIG. **10**, the reflux component **23** further includes multiple fixing protrusions **233**. Each of the fixing protrusions **233** is disposed on a side of a corresponding reflux blade **232** away from the partition plate **231** and being connected to the corresponding reflux blade **232**. An end of the fixing protrusion **233** away from the corresponding reflux blade **232** is fixedly connected to the impeller assembly housing **21** to ensure the stability of the partition plate **231** and the reflux blades **232**. The number of the fixing protrusions **233** is less than the number of the reflux blades **232**, for example, the number of the fixing protrusions **233** is 4, and the number of the reflux blades **232** is 16.

[0122] In some embodiments, the fixing protrusions **233** may be fixedly connected to the partition plate **231** and the impeller assembly housing **21**, respectively, but not to the reflux blades **232**. The number of the fixing protrusions **233** may be greater than or equal to the number of the reflux blades **232**. No limitations are placed on the number and distribution of the fixing protrusions **233**.

[0123] In some embodiments, as shown in FIG. **9**, the impeller assembly **2** further includes an elastic sealing ring **25**, and the partition plate **231** of the reflux component **23** defines a mounting cavity **231A**. The elastic sealing ring **25** and the output shaft **11** are disposed in the mounting cavity **231A**. The elastic sealing ring **25** is sleeved on the output shaft **11** and is disposed between the output shaft **11** and the reflux component **23**. The elastic sealing ring **25** is fixedly connected to the reflux component **23** and rotatably connected to the output shaft **11**.

[0124] FIG. **11** is a structural schematic view of an elastic sealing ring according to some embodiments of the present disclosure, and FIG. **12** is a cross-sectional schematic view of FIG. **11** along line A-A. referring to FIGS. **9**, **11**, and **12**, the elastic sealing ring **25** may include a sealing ring body **251** and an elastic sealing tooth **252**. Each of the sealing ring body **251** and the elastic sealing tooth **252** has a ring-shaped structure, and both the sealing ring body **251** and the elastic sealing tooth **252** are sleeved on the output shaft **11**. The sealing ring body **251** is disposed between the reflux component **23** and the elastic sealing tooth **252** and is fixedly connected to the reflux component **23** and the elastic sealing tooth **252**, respectively. The elastic sealing tooth **252** is disposed between the sealing ring body **251** and the output shaft **11**, and the elastic sealing tooth **252** is fixedly connected to the sealing ring body **251** while being rotatably connected to the output shaft **11**.

[0125] The sealing ring body **251** may be fixedly connected to the reflux component **23** by welding, gluing, snap-fitting, or plug-fitting. The sealing ring body **251** and the reflux component **23** may be detachably connected to facilitate assembly between the sealing ring body **251** and the reflux component **23**. Moreover, the detachable connection may facilitate the replacement of the elastic sealing ring **25** in the event of damage to the sealing ring body **251** or the elastic sealing tooth **252**. The sealing ring body **251** and the elastic sealing tooth **252** may be integrally formed, or they may be fixedly connected by snap-fitting or gluing.

[0126] In some embodiments, the elastic sealing tooth **252** and the output shaft **11** may be fitted with a clearance to ensure that the position of the reflux component **23** in the impeller assembly **2** remains unchanged during high-speed rotation of the output shaft **11**, i.e. the reflux component **23** does not rotate with the output shaft **11**.

[0127] In some embodiments, the elastic sealing tooth **252** and the output shaft **11** may be connected by a bearing rotation. In this case, sealing rings are arranged on both ends of the bearing perpendicular to the axis *m* of the output shaft **11**. The sealing rings seal the bearing, thereby preventing lubricant from overflowing from the bearing and further preventing gas in the reflux channel or the second flow channel from flowing back to the first flow channel through the bearing.

[0128] In some embodiments, any of the elastic sealing rings **25** provided by the embodiments of the present disclosure may be made of a soft rubber material, such as nitrile rubber, silicone rubber, polyethylene or polypropylene. The material of the elastic sealing ring **25** is only described here as an example and is not limited in any way. Moreover, the dimensions of the elastic sealing ring **25** may be determined according to the actual product requirements, which is not limited herein.

[0129] In some embodiments, referring to FIG. **9**, the impeller assembly **2** may further include a sealing cotton **26**. The sealing cotton **26** is arranged between the impeller assembly housing **21** and the first impeller **22** and/or between the impeller assembly housing **21** and the second impeller **24**, thereby preventing the backflow of gas through the gap between the first impeller **22** and the impeller assembly housing **21** and the backflow of gas through the gap between the second impeller **24** and the impeller assembly housing **21**. This solution may further improve the tightness of the impeller assembly **2**, thereby further improving the operating efficiency of the electric fan.

[0130] FIG. **13** is a partial structural schematic view of an electric fan according to some embodiments of the present disclosure. As an example, as shown in FIG. **13**, the impeller assembly **2** further includes a sealing bearing **27**. The sealing bearing **27** is sleeved on the output shaft **11** and is disposed between the elastic sealing ring **25** and the output shaft **11**. The seal bearing **27** is fixedly connected to the output shaft **11** and the elastic sealing ring **25**, respectively. The seal bearing **27** may be connected to the output shaft **11** by means of a press fit or a key connection, and the seal bearing **27** may be connected to the elastic sealing ring **25** by means of a press fit, gluing, plugging, or welding. Sealing rings are arranged on both ends of the seal bearing **27** perpendicular to the axis *m* of the output shaft **11**. The sealing rings seal the seal bearing **27**, thereby preventing lubricant from overflowing from the seal bearing **27** and further preventing gas in the reflux channel or the second flow channel from flowing back to the first flow channel through the seal bearing **27**.

[0131] This solution may ensure both a rotational connection between the reflux component **23** and the output shaft **11** and a good tightness between the reflux component **23** and the output shaft **11**. In addition, the elastic sealing ring **25** may provide a certain vibration-reducing effect on the output shaft **11**, thereby not only improving the working efficiency of the electric blower, but also improving the service life of the electric blower.

[0132] FIG. **14** is a partial structural schematic view of a diffuser assembly according to some embodiments of the present disclosure. Referring to FIGS. **9** and **14**, the diffuser assembly **3** includes a first axial flow diffuser **31**, the first axial flow diffuser **31** including a first diffuser impeller **311** and a first diffuser housing **312**. The first diffuser impeller **311** includes a first inner shell **3111** and multiple first diffuser blades **3112**.

[0133] Referring to FIGS. **9** and **14**, the first inner shell **3111** may have a ring-shaped structure for being sleeved outside the output shaft **11**. The first inner shell **3111** is disposed on a side of the impeller assembly **2** close to the stator-rotor component **13**, and the first inner shell **3111** is sealed and connected to the impeller assembly **2**. In particular, the first inner shell **3111** and the impeller assembly **2** may be connected to each other by welding. As an example, the first inner shell **3111** has a circular ring structure, e.g., the first inner shell **3111** has a circular ring shape in the radial cross-section, and there are no limitations on the cross-section shape of the ring structure.

[0134] Referring to FIGS. **9** and **14**, taking the first inner shell **3111** with a circular annular structure as an example, the multiple first diffuser blades **3112** are distributed along a circumferential direction of an outer wall of the first inner shell **3111**, and each first diffuser blade

3112 is connected to the first inner shell **3111**. The first diffuser blade **3112** and the first inner shell **3111** may be integrally formed, or they may be connected by welding, snap-fitting, or gluing. [0135] Referring to FIG. 9, the first diffuser housing **312** has a ring-shaped structure, and the first diffuser housing **312** is sleeved on the multiple first diffuser blades **3112**. The first diffuser housing **312** and the multiple first diffuser blades **3112** may be sealed and connected by means of one-piece forming, welding or adhesive bonding, etc., or a certain installation gap may be left between the first diffuser housing **312** and the multiple first diffuser blades **3112**. For example, a gap of 0.5 mm (or 0.2 mm, 1 mm, etc.) is defined between an inner wall of the first diffuser housing **312** and a surface of the first diffuser blade **3112** away from the first inner shell **3111**. The first diffuser housing **312** is sealed to the impeller assembly **2** to prevent gas leakage.

[0136] In the first diffuser assembly **3**, a first diffuser flow channel is formed between the first diffuser impeller **311** and the inner wall of the first diffuser housing **312**, and the first diffuser flow channel is in communication with the flow channel in the impeller assembly **2**. Since the first inner shell **3111** and the first diffuser housing **312** are respectively sealed and connected to the impeller assembly **2**, during the operation of the electric fan, the gas first enters the flow channel in the impeller assembly **2**, and after being accelerated by the impeller assembly **2**, it flows into the first diffuser flow channel for primary pressure-expansion and deceleration treatment. By virtue of this solution, the electric fan may be ensured to operate normally and its work efficiency may be ensured.

[0137] The solutions provided in the embodiments of the present disclosure may be that in the first axial flow diffuser **31**, the first diffuser blades **3112** are unevenly distributed along the circumferential direction of the outer wall of the first inner shell **3111**. That is, in the first axial flow diffuser **31**, the flow channel formed between two adjacent first diffuser blades **3112** has different sizes in the circumferential direction of the outer wall of the first inner shell **3111**. Since the frequency of the gas flow is not the same in flow channels of different widths (i.e. different sizes), the pressure pulsations generated on both sides of the first diffuser blade **3112** do not superimpose on each other, and may even cancel each other out, which may help to reduce the noise generated by the diffuser assembly **3** and the operating noise of the electric fan during operation.

[0138] Referring to FIG. 9, the diffuser assembly **3** may further include a second axial flow diffuser **32**, the second axial flow diffuser **32** including a second diffuser impeller **321** and a second diffuser housing **322**. The second diffuser impeller **321** includes a second inner shell **3211** and multiple second diffuser blades **3212**.

[0139] Referring to FIGS. 9 and 14, the second inner shell **3211** has a ring-shaped structure, and the second inner shell **3211** is sleeved on the output shaft **11**. The second inner shell **3211** is disposed on a side of the first inner shell **3111** away from the impeller assembly **2** and is sealed and connected to the first inner shell **3111**. The second inner shell **3211** and the first inner shell **3111** may be integrally formed, or they may be connected by means of a screw, welding, or riveting connection. In particular, the second inner shell **3211** and the first inner shell **3111** may be detachably connected, which is conducive to reducing the difficulty of assembling the diffuser assembly **3**, thereby improving production efficiency and facilitating subsequent maintenance.

[0140] Referring to FIGS. 9 and 14, taking the second inner shell **3211** with a circular annular structure as an example, the multiple second diffuser blades **3212** are disposed on the outer wall of the second inner shell **3211** and distributed along a circumferential direction of the outer wall of the second inner shell **3211**, and the multiple second diffuser blades **3212** are respectively connected to the second inner shell **3211**. The second diffuser blade **3212** and the second diffuser blade **3212** may be integrally formed, or they may be connected by welding, snap-fitting, plug-in, or gluing.

[0141] Referring to FIG. 9, the second diffuser housing **322** has a ring-shaped structure, and the second diffuser housing **322** is sleeved on the multiple second diffuser blades **3212**. The second diffuser housing **322** and the first diffuser housing **312** may be sealed and connected. The second diffuser housing **322** and the first diffuser housing **312** may be connected by threaded connection,

welding, snap-fitting, etc., or they may be integrally formed. In particular, the first diffuser housing **312** and the second diffuser housing **322** may be connected in a detachable manner, which is conducive to reducing the difficulty of assembling the diffuser assembly **3**, thereby improving production efficiency and also facilitating subsequent maintenance. The second diffuser housing **322** and the second diffuser blade **3212** may be fixedly connected by means of one-piece forming, welding, or gluing, or a gap may be left between the second diffuser housing **322** and the second diffuser blade **3212**. For example, a gap of 0.5 mm (or 0.2 mm, 1 mm, etc.) is defined between an inner wall of the second diffuser housing **322** and a surface of the second diffuser blade **3212** away from the second inner shell **3211**.

[0142] In the diffuser assembly **3**, a second diffuser flow channel is formed between the second diffuser impeller **321** and the inner wall of the second diffuser housing **322**, and the second diffuser flow channel is in communication with the first diffuser flow channel. During operation of the electric fan, the gas flows into the first diffuser flow channel first, undergoes the primary pressure-expansion and deceleration treatment, and then enters the second diffuser flow channel to undergo a secondary pressure-expansion and deceleration treatment. After undergoing the two pressure-expansion and deceleration treatments, the gas flows to the outside of the diffuser assembly **3**.

[0143] In some embodiments, the diffuser assembly **3** may include more than two axial flow diffusers. The solution of using a multi-stage axial flow diffuser is more conducive to improving the pressure-expansion of the electric fan to meet product development needs. When the diffuser assembly **3** includes more than two axial flow diffusers, the two or more axial flow diffusers are coaxially arranged and connected in sequence, and it is only necessary to ensure that the blades in the adjacent two axial flow diffusers are distributed in a misaligned manner. The specific distribution method of the blades, as well as the connection relationship between the multiple axial flow diffusers, is similar to the above situation with the first axial flow diffuser **31** and the second axial flow diffuser **32**, so it will not be described in detail herein.

[0144] The multi-stage axial flow diffuser may be advantageous for improving the pressure-expansion and deceleration capabilities of the electric fan to meet product development needs, and further advantageous for reducing the radial dimensions of the electric fan with the same pressure-expansion capability, thereby improving the applicability of the electric fan.

[0145] FIG. **15** is a cross-sectional structural schematic view of an electric fan according to some embodiments of the present disclosure. Referring to FIG. **15**, a first end surface **12B** is formed on an end of the bracket **12** adjacent to the diffuser assembly **3**, and a gap is defined between the air outlet **3A** of the diffuser assembly **3** and the first end surface **12B** in the extension direction of the axis *m* of the output shaft **11**. A deflector **4** is arranged on the first end surface, e.g., the deflector **4** is disposed on and connected to the first end surface **12B** of the bracket **12**. Furthermore, at least a portion of the deflector **4** is opposite the air outlet **3A** to adjust the flow direction of the gas exiting the air outlet **3A**. In particular, at least a portion of the deflector **4** is opposite the air outlet **3A**, and it may be considered that, in the extension direction of the axis *m* of the output shaft **11**, a positive projection of the deflector **4** at least partially overlaps with a positive projection of the air outlet **3A**.

[0146] In some embodiments, the positive projection of the deflector **4** completely overlaps with the positive projection of the air outlet **3A**. In other embodiments, the positive projection of the deflector **4** is disposed within the positive projection of the air outlet **3A**, i.e. the positive projection of the deflector **4** partially overlaps with the positive projection of the air outlet **3A**. In other embodiments, the positive projection of the air outlet **3A** is disposed within the positive projection of the deflector **4**, i.e. the positive projection of the air outlet **3A** partially overlaps with the positive projection of the deflector **4**. In other embodiments, a portion of the positive projection of air outlet **3A** overlaps with a portion of the positive projection of deflector **4**.

[0147] In the related art, in an electric fan in which the motor is arranged downstream of the diffuser, the gas flowing out of the diffuser blows towards the motor approximately vertically, and

after the gas flows impinges on the motor, it is forced to turn vertically and split, resulting in large aerodynamic losses. Moreover, the gas impinging on the motor generates large impact noise, thereby increasing the operating noise when the electric fan is in operation.

[0148] In the solutions provided in the embodiments of the present disclosure, the deflector **4** is arranged between the diffuser assembly **3** and the bracket **12**, and at least a portion of the deflector **4** is opposite the air outlet **3A** of the diffuser assembly **3**. The deflector **4** can adjust the flow direction of the gas. Therefore, under the action of the deflector **4**, the gas flowing out of the air outlet **3A** of the diffuser assembly **3** is gradually guided to make turns and split flows, and will no longer blow vertically towards the first end surface **12B** and the stator-rotor component **13** (e.g., will no longer blow vertically towards the motor assembly **1**), which may be conducive to reducing the impact of the airflow on the motor assembly **1** to eliminate the impact noise generated by the impact of the airflow on the motor assembly **1**, and furthermore, conducive to reducing the operating noise of the electric fan during operation.

[0149] FIG. **16** is a structural schematic view of a deflector according to some embodiments of the present disclosure. Referring to FIGS. **15** and **16**, the deflector **4** includes a deflection portion **41**, and the deflection portion **41** is configured to adjust a flow direction of gas exiting the air outlet **3A**. As an example, the deflection portion **41** is a raised structure, and the raised structure (e.g., the deflection portion **41**) protrudes toward the air outlet **3A**. In other words, in a first direction **O**, a radial width of the deflection portion **41** gradually decreases; where the first direction **O** is parallel to the extension direction of the axis **m** of the output shaft **11** and points from the first end surface **12B** towards the air outlet **3A**, and the radial width is a width of the deflection portion **41** perpendicular to the first direction **O**. In some embodiments, a point on the deflection portion **41** furthest from the first end surface **12B** is opposite the air outlet **3A**, e.g., in the extension direction of the axis **m** of the output shaft **11**, a positive projection of the point on the deflection portion **41** furthest from the first end surface **12B** is disposed within the positive projection of the air outlet **3A**.

[0150] The closer a portion of the deflection portion **41** is to the air outlet **3A**, the smaller the radial dimension of the portion is, i.e. the portion of the deflection portion **41** that is close to the air outlet **3A** is more acute than another portion that is further away from the air outlet. In general, an acute part can produce a better air diversion effect. Therefore, this solution is conducive to improving the gas-diversion and-guidance effect of the deflection portion **41** of the deflector **4**, thereby further reducing aerodynamic losses and reducing the operating noise of the electric fan.

[0151] As an example, when the radial width of the deflection portion **41** gradually decreases in the first direction **O**, it may be considered that two sides of the deflection portion **41** perpendicular to the first direction **O** are smooth surfaces, which may reduce wind resistance, thereby reducing aerodynamic losses when the airflow flows past both sides of the deflection portion **41** to improve gas flow efficiency.

[0152] In some embodiments, the point on the deflection portion **41** furthest from the first end surface **12B** is disposed within the diffuser assembly **3** in the extension direction of the axis **m** of the output shaft **11**, such that the diversion action is brought forward, thereby further reducing aerodynamic losses and improving gas flow efficiency. However, the greater the distance between the point on the deflection portion **41** furthest from the first end surface **12B** and the diffuser assembly **3**, the closer the point is to the impeller assembly **2**, in which case the flow rate of the gas is faster, and the impact on the deflection portion **41** is greater. Although the gas flow efficiency may be improved, the impact of the gas on the deflection portion **41** is likely to increase the wear of the deflection portion **41** and generate some noise.

[0153] Therefore, in some embodiments, a gap is defined between the point on the deflection portion **41** furthest from the first end surface **12B** and the air outlet **3A** in the extension direction of the axis **m** of the output shaft **11**. The gap may be determined according to the actual product requirements, for example, the gap may be half the distance between the first end surface **12B** and

the air outlet **3A**. No limit is placed on the gap between the point on the deflection portion **41** furthest from the first end surface **12B** and the air outlet **3A**. By virtue of this solution, after the gas flows out of the air outlet **3A**, it diffuses and slows down to a certain extent within this gap. The impact of the diffused and slowed down gas on the deflection portion **41** may be significantly reduced, which is beneficial for improving the efficiency of gas flow, reducing wear on the deflection portion **41**, and reducing the noise generated by the gas impacting the deflection portion **41**.

[0154] Referring to FIGS. **15** and **16**, the deflection portion **41** protrudes towards the air outlet **3A**, and a surface of the deflection portion **41** close to the air outlet **3A** is curved. In other words, in a radial cross-section of the electric fan, the surface of the deflection portion **41** close to the air outlet **3A** is a circular arc.

[0155] In some embodiments, the curved surface is an arc of equal radius, e.g., the radii corresponding to any point on the curved surface are equal. In this case, the radius of the curved surface may be considered to be the axial length of the deflection portion **41** in the extension direction of the axis **m** of the output shaft **11**. In other words, in a radial cross-section of the electric fan, the circular arc formed by the surface of the deflection portion **41** close to the air outlet **3A** is a semicircle.

[0156] As an example, when the stator **132** is sleeved outside the rotor **131**, and the air outlet **3A** is opposite the first end surface **12B** and a portion of the stator **132**, the radius of the curved surface may be equal to half the radial distance from an outer wall of a stator winding in the stator **132** to the outer wall of the bracket. In some embodiments, the radius of the curved surface may be equal to the radial dimension of the air outlet **3A**, or equal to half the radial dimension of the deflection portion **41**. No limitations are placed on the axial length of the deflection portion **41**.

[0157] In other embodiments, the curved surface described above is a curved surface with a variable radius. As an example, referring to FIG. **15**, the radius corresponding to any point on the curved surface is negatively correlated with the distance from any point to the axis **m**. In other words, in the radial cross-section of the electric fan, the arc formed by the surface of the deflection portion **41** close to the air outlet **3A** is a variable radius arc; on the variable radius arc, the further away a point is from the axis **m**, the less the radius corresponding to the point, and the closer the point is to the axis **m**, the greater the radius corresponding to the point. In some embodiments, the radius corresponding to any point on the curved surface may be positively correlated with the distance from any point to the axis **m**.

[0158] As an example, when the stator **132** is sleeved outside the rotor **131**, and the air outlet **3A** is opposite to the first end surface **12B** and a portion of the stator **132**, an average radius of the curved surface with variable radius may be equal to half the radial distance between the outer wall of the stator winding in the stator **132** and the outer wall of the bracket, or it may be equal to the radial dimension of the air outlet **3A**, or it may be equal to half the radial dimension of the deflection portion **41**.

[0159] FIG. **17** is a cross-sectional structural schematic view of an electric fan according to some embodiments of the present disclosure. As an example, as shown in FIG. **17**, a radial cross-section of the deflection portion **41** is triangular, that is, the deflection portion **41** is triangular in the radial cross-section of the electric fan. As an example, the radial cross-section of the deflection portion **41** is an equilateral triangle, where the height of the equilateral triangle may be equal to half the radial distance between the outer wall of the stator winding in the stator **132** and the outer wall of the bracket, or it may be equal to the radial dimension of the air outlet **3A**, or it may be equal to half the radial dimension of the deflection portion **41**. When the radial cross-section of the deflection portion **41** is triangular, the shape and dimensions of the triangle may be determined according to the actual product requirements, which is not limited herein.

[0160] In some embodiments, referring to FIGS. **15** and **16**, the deflector **4** may further include a fixing portion **42**. The fixing portion **42** is disposed on a side of the deflection portion **41** away

from the diffuser assembly **3** and is sleeved radially outside the bracket **12**, where the radial direction of the bracket **12** is perpendicular to the direction of the axis **m** of the output shaft **11**. The fixing portion **42** is connected to the flow guiding portion **41** and the bracket **12**, respectively. The fixing portion **42** and the flow guiding portion **41** may be integrally formed, or they may be fixed by welding, riveting, or gluing, etc. The fixing portion **42** and the bracket **12** may be integrally formed, or they may be fixed by welding, snap-fitting, etc.

[0161] In some embodiments, the deflection portion **41** and the fixing portion **42** are integrally formed, and the fixing portion **42** is welded to the bracket **12**. In this case, the deflection portion **41** and the bracket **12** may be fixedly connected or only in contact with each other without being connected.

[0162] In some embodiments, the deflection portion **41**, the fixing portion **42**, and the bracket **12** may be integrally formed, thereby facilitating the integration of components. No limitation is made here on the connection method between the deflection portion **41**, the fixing portion **42**, and the bracket **12**.

[0163] FIG. **18** is a partial structural schematic view of an electric fan according to some embodiments of the present disclosure. Referring to FIGS. **15** and **18**, when the support member **122** includes multiple support arms **1221**, the electric fan may include multiple any one of the above-mentioned deflectors **4**, and a deflector **4** is arranged between each adjacent two support arms **1221**.

[0164] As an example, referring to FIG. **18**, the multiple support arms **1221** are distributed at equal intervals along the circumferential direction of the output shaft **11**, in which case the multiple deflection portions **4** are also distributed at equal intervals along the circumferential direction of the output shaft **11**. In some embodiments, the multiple support arms **1221** may be distributed at non-equal intervals along the circumferential direction of the output shaft **11**. In this case, it is only necessary to arrange the deflection portion **4** between each adjacent two support arms **1221**. There are no limitations on the number and distribution of the deflection portions **4**.

[0165] FIG. **19** is a partial structural schematic view of an electric fan according to some embodiments of the present disclosure. As an example, as shown in FIG. **19**, when the bracket **12** includes multiple support arms **1221**, the deflector **4** may include multiple deflection portions **41** and one fixing portion **42**. Each of the multiple deflection portions **41** is disposed between corresponding two adjacent support arms **1221** and connected to the fixing portion **42**.

[0166] In some embodiments, the deflector **4** may include multiple connecting portions **43**. As shown in FIG. **19**, the connecting portions **43** correspond to the support arms **1221** in a one-to-one correspondence, and the connecting portion **43** is disposed on an outer side of the support arm **1221** and between corresponding two adjacent deflection portions **41**. The connecting portion **43** may be connected to the deflection portion **41** and the fixing portion **42**, respectively, to improve the strength and rigidity of the entire deflector **4**. In some embodiments, the connecting portion **43** may further be connected to the support arm **1221**, thereby improving the stability of the deflector **4** when mounted on the bracket **12**.

[0167] In the solutions provided in the present embodiments, the deflector **4** is arranged between the diffuser assembly **3** and the bracket **12**, and at least a portion of the deflector **4** is opposite the air outlet **3A** of the diffuser assembly **3**. The deflector **4** can adjust the direction of the gas flow, such that under the action of the deflector **4**, the gas flowing out of the air outlet **3A** of the diffuser assembly **3** is gradually guided to make turns and split flows, and will no longer blow vertically towards the first end surface **12B** and the stator-rotor component **13** (e.g., will no longer blow vertically towards the motor assembly **1**), which may be conducive to reducing the impact of the airflow on the motor assembly **1** to eliminate the impact noise generated by the impact of the airflow on the motor assembly **1**, and furthermore, conducive to reducing the operating noise of the electric fan during operation.

[0168] FIG. **20** is a cross-sectional structural schematic view of an electric fan according to some

embodiments of the present disclosure. Referring to FIG. 20, the electric fan may further include a mesh structure 5, which is disposed on a side of the diffuser assembly 3 away from the impeller assembly 2 and is connected to the diffuser assembly 3. The gas flowing out of the air outlet 3A of the diffuser assembly 3 encounters the mesh structure 5, and the flow rate and direction are adjusted under the action of the mesh structure 5. As an example, as shown in FIG. 20, the diffuser assembly 3 and the stator-rotor component 13 are spaced apart along the extension direction of the axis m of the output shaft 11. In this case, the mesh structure 5 covers the gap between the diffuser assembly 3 and the stator-rotor component 13, and the mesh structure 5 is respectively connected to the diffuser assembly 3 and the bracket 12.

[0169] The mesh structure 5 can adjust the flow rate distribution of the gas at the air outlet 3A of the diffuser assembly 3, such that the flow rate distribution of the gas at the air outlet 3A is uniform. Furthermore, the mesh structure 5 has an effect of slowing down and adjusting the flow direction of the gas, thereby reducing the impact of the gas on a surface of the motor or a surface of an electronic control board, thereby reducing or even eliminating impact noise, which may reduce the operating noise of the electric fan when in operation.

[0170] FIG. 21 is a schematic diagram of gas flow, when there is no mesh structure, according to some embodiments of the present disclosure, and FIG. 22 is a schematic diagram of gas flow, when there is a mesh structure, according to some embodiments of the present disclosure.

[0171] As shown in FIG. 21, in an electric fan without the mesh structure 5, when the gap between the diffuser assembly 3 and the stator-rotor component 13 is equal to 6 mm, the gas flowing out of the air outlet 3A of the diffuser assembly 3 will impact the end surface of the stator-rotor component 13 at a high speed, and the gas will only spread to both sides (mainly to a side away from the output shaft 11) after reaching the end surface of the stator-rotor component 13. In this case, the bending angle of the gas flow during diffusion is greater than or equal to 60 degrees, or even up to 90 degrees.

[0172] As shown in FIG. 22, in an electric fan with the mesh structure 5, when the gap between the diffuser assembly 3 and the stator-rotor component 13 is equal to 6 mm, the gas flowing out of the air outlet 3A of the diffuser assembly 3 may be gradually decelerated under the action of the mesh structure 5, and under the action of the mesh structure 5, the gas will gradually spread to the side away from the output shaft 11 before reaching the end surface of the stator-rotor component 13. In this case, the bending angle of the gas flow during diffusion is less than or equal to 45 degrees.

[0173] In some embodiments, when the stator 132 is sleeved outside the rotor 131, the gap between the diffuser assembly 3 and the stator-rotor component 13 may be equal to or more than twice the radial thickness of the stator 132. In some embodiments, when the stator 132 and rotor 131 are coaxially arranged along the axis of the output shaft, the gap between the diffuser assembly 3 and the stator-rotor component 13 may be equal to the radius of the stator 132. The above is only some examples of the gap between the diffuser assembly 3 and the stator-rotor component 13. The gap between the diffuser assembly 3 and the stator-rotor component 13 may be set according to the gas flow rate discharged from the diffuser assembly 3 in the actual product, and no limit is imposed herein.

[0174] As can be seen in FIGS. 21 and 22, by virtue of the proposed solutions, the gas flowing out of the air outlet 3A of the diffuser assembly 3 may have its flow rate reduced before reaching the stator-rotor component 13 under the action of the mesh structure 5. Furthermore, the flow direction may be adjusted under the action of the mesh structure 5, thereby reducing the impact of the gas on the stator-rotor component 13 and thus reducing impact noise and the operating noise of the electric fan.

[0175] As an example, a length of the mesh structure 5 in the extension direction of the axis m of the output shaft 11 may be greater than the gap between the stator-rotor component 13 and the diffuser assembly 3. As shown in FIG. 20, two ends of the mesh structure 5 may respectively wrap around the outer wall of the diffuser assembly 3 and the outer wall of the bracket 12. The mesh

structure **5** and the outer wall of the diffuser assembly **3** may be fixedly connected by welding, gluing, or riveting, and the mesh structure **5** and the outer wall of the bracket **12** may be fixedly connected by welding, gluing or riveting, thereby improving the stability of the mesh structure **5** in the electric fan. In some embodiments, the mesh structure **5** and the outer wall of the diffuser assembly **3** may be detachably connected by means of a snap fit, a plug-in connection or a threaded connection, and the mesh structure **5** and the outer wall of the bracket **12** may be detachably connected by means of a snap fit, a plug-in connection or a threaded connection, thereby facilitating the replacement of the mesh structure **5**.

[0176] In some embodiments, an end of the mesh structure **5** is connected to an end surface where the air outlet **3A** is disposed, and the other end of the mesh structure **5** wraps around the outer wall of the bracket **12**. Or, an end of the mesh structure **5** wraps around the outer wall of the diffuser assembly **3**, and the other end of the mesh structure **5** is connected to an end surface of the bracket **12** close to the diffuser assembly **3**. Or, the two ends of the mesh structure **5** are respectively connected to the end surface where the air outlet **3A** is disposed and the end surface of the bracket **12** close to the diffuser assembly **3**.

[0177] FIG. **23** is a structural schematic view of an electric fan according to some embodiments of the present disclosure, and FIG. **24** is a cross-sectional structural schematic view of an electric fan according to some embodiments of the present disclosure. As an example, as shown in FIG. **24**, the motor assembly **1** further includes an electronic control board **18**. The stator-rotor component **13** is disposed on an inner side of the diffuser assembly **3**, that is, the diffuser assembly **3** is sleeved on the bracket **12**, and an inner wall of the diffuser assembly **3** is connected to the outer wall of the bracket **12**. The electronic control board **18** is disposed on a side of the mesh structure **5** away from the diffuser assembly **3**, the electronic control board **18** is opposite the air outlet **3A** of the diffuser assembly **3**, and the electronic control board **18** is respectively connected to the mesh structure **5** and the bracket **12**. In other words, the electronic control board **18** is disposed on a side of the bracket **12** away from the impeller assembly **2**, and a gap is defined between the air outlet **3A** and the electronic control board **18** in the extension direction of the axis **m** of the output shaft **11**. The mesh structure **5** covers the gap. The electronic control board **18** may further be electrically connected to the stator-rotor component **13** to control the rotation of the rotor **131** in the stator-rotor component **13**, thereby driving the impeller in the impeller assembly **2**.

[0178] When the stator **132** is sleeved outside the rotor **131**, the gap between the diffuser assembly **3** and the electronic control board **18** may be equal to or greater than twice the radial thickness of the stator **132**. In some embodiments, when the stator **132** and rotor **131** are arranged coaxially along the axis of the output shaft, the gap between the diffuser assembly **3** and the electronic control board **18** may be equal to the radius of the stator **132**. The above is only some examples of the gap between the diffuser assembly **3** and the electronic control board **18**. The gap between the diffuser assembly **3** and the electronic control board **18** may be set according to the gas flow rate discharged from the diffuser assembly **3** in the actual product, and no limit is imposed herein.

[0179] In the electric fan provided in the embodiments of the present disclosure, the diffuser assembly **3** is sleeved outside the bracket **12**, and it may be considered that the diffuser assembly **3** is sleeved outside the entire motor assembly **1**, which is conducive to reducing the length of the electric fan in the extension direction of the axis **m** of the output shaft **11**. Moreover, the gas flowing out of the air outlet **3A** of the diffuser assembly **3** may reduce the flow rate and adjust the flow direction under the action of the mesh structure **5** before reaching the electronic control board **18**, thereby reducing the impact of the gas on the stator-rotor component **13**, and thus reducing the impact noise and the operating noise of the electric fan.

[0180] Referring to FIGS. **20** and **24**, the porosity of the mesh structure **5** provided in the embodiments of the present disclosure may range from 60% to 85%. In other words, on an outer wall of the mesh structure **5** (e.g., a surface of the mesh structure **5** away from the output shaft **11**), the total area occupied by the through holes accounts for 60% to 85% of the total area of the outer

wall of the mesh structure **5**. No limit is imposed on the porosity of the mesh structure **5**.

[0181] Based on the same technical concept, the present disclosure provides a terminal device. FIG. **25** is a structural schematic view of a terminal device according to some embodiments of the present disclosure, and the terminal device includes any one of the electric fans **01** provided by the present disclosure. The terminal device may be a cleaning device, such as a vacuum cleaner or a sweeper.

[0182] As an example, the terminal device is a vacuum cleaner. Referring to FIG. **25**, the vacuum cleaner may further include an air intake apparatus **02**, a dust collection chamber **03**, and an exhaust pipe **04**. The air intake apparatus **02**, the dust collection chamber **03**, the electric fan **01**, and the exhaust pipe **04** are connected in sequence, where an end of the air intake apparatus **02** away from the dust collection chamber **03** (e.g., an end of the air intake apparatus **02** not connected to the dust collection chamber **03**) is connected to outside, and an end of the exhaust pipe **04** away from the electric fan **01** (e.g., an end of the exhaust pipe **04** not connected to the electric fan **01**) is connected to outside.

[0183] When the vacuum cleaner is in operation, external air and target objects (e.g. dust, spilled water, dropped food, etc.) are sucked into the air intake apparatus **02** by the electric fan **01**, and the target objects then enter the dust collection chamber **03** together with the air. The dust collection chamber **03** may be arranged with a filter apparatus, and the dust collection chamber **03** filters the target objects and air. The target objects are retained in the dust collection chamber **03**, and the filtered gas enters the electric fan **01**, is accelerated and pressure-expanded, and then flows out of the diffuser assembly **3**. The outgoing gas passes through the outer surface and/or the interior of the stator-rotor component **13**, taking away the heat generated by the stator-rotor component **13** and flowing to the exhaust pipe **04**, and finally into the atmosphere. After the vacuum cleaner has completed the cleaning function on the target objects, the gas flowing out of the diffuser assembly **3** is fully utilized to dissipate heat from the stator-rotor component **13**, which is beneficial to improving the heat dissipation effect of the stator-rotor component **13**, thereby improving the work reliability and service life of the entire electric fan (and vacuum cleaner).

[0184] Moreover, in the relevant art, the radial dimension of the electric fan is affected by the diffuser assembly **3**. Since the diffuser assembly **3** is sleeved outside the motor, the radial dimension of the entire electric fan is usually large. In the proposed solutions, the stator-rotor component **13** is disposed outside the diffuser assembly **3**, which reduces the radial dimension of the diffuser assembly **3** and thus the radial dimension of the entire electric motor, which is conducive to the application of the electric motor in more products, thereby improving the applicability of the electric motor.

[0185] The above is only some implementations of the present disclosure and is not intended to limit the present disclosure. Any modifications, equivalent replacements, and improvements made within the spirit and principles of the present disclosure shall be included in the scope of the present disclosure.

Claims

1. An electric fan, comprising: a motor assembly; an impeller assembly; and a diffuser assembly, wherein the motor assembly comprises an output shaft, a bracket, and a stator-rotor component; the bracket and the stator-rotor component are both sleeved on the output shaft; the bracket has an accommodation region, and the stator-rotor component is disposed in the accommodation region; the impeller assembly is disposed on a side of the stator-rotor component and is sleeved on the output shaft; the diffuser assembly is sleeved on the output shaft and is disposed between the stator-rotor component and the impeller assembly; and wherein a part of gas passing through the diffuser assembly is configured to flow toward an outside of the bracket, and another part of the gas is configured to flow toward an interior of the stator-rotor component.

2. The electric fan according to claim 1, wherein a gap is defined between the stator-rotor component and the diffuser assembly in an extension direction of an axis of the output shaft.
3. The electric fan according to claim 2, wherein the diffuser assembly has an air outlet on an end close to the stator-rotor component; in the extension direction of the axis of the output shaft, a positive projection of the air outlet at least partially overlaps with a positive projection of the stator-rotor component.
4. The electric fan according to claim 3, wherein the stator-rotor component comprises a rotor and a stator; the rotor is sleeved on the output shaft, and the stator is sleeved outside the rotor; in the extension direction of the axis of the output shaft, a positive projection of the air outlet at least partially overlaps with a positive projection of the stator.
5. The electric fan according to claim 4, wherein in the extension direction of the axis of the output shaft, the positive projection of the air outlet is a first projection, and the positive projection of the stator is a second projection; an area where the first projection and the second projection overlap is greater than or equal to half an area of a small projection; the small projection is one of the first projection and the second projection, and the area of the small projection is less than an area of the other of the first projection and the second projection.
6. The electric fan according to claim 1, wherein the bracket comprises a bracket body and a support member; the bracket body is sleeved on the output shaft; the support member is disposed on a side of the bracket body close to the diffuser assembly, and the support member is connected to the diffuser assembly and the bracket body, respectively.
7. The electric fan according to claim 6, wherein the motor assembly further comprises a connecting member; the connecting member is sleeved on the output shaft and is disposed on a side of the support member away from the bracket body; the connecting member is disposed on a radial inner side of the diffuser assembly; the connecting member is connected to the support member and the diffuser assembly, respectively.
8. The electric fan according to claim 6, wherein the support member comprises at least one support arm and a support connection; the at least one support arm is disposed on a side of the diffuser assembly close to the bracket body; the support connection is disposed on a side of the at least one support arm away from the diffuser assembly, and the support connection is connected to the at least one support arm and the bracket body, respectively.
9. The electric fan according to claim 8, wherein the at least one support arm is recessed toward the output shaft.
10. The electric fan according to claim 8, wherein the at least one support arm is a plurality of support arms, and the plurality of support arms are distributed at equal intervals along a circumferential direction of the output shaft.
11. The electric fan according to claim 7, wherein the motor assembly further comprises a first bearing and a second bearing; the connecting member has a first bearing chamber, and the bracket body has a second bearing chamber; the first bearing and the second bearing are both sleeved on the output shaft, with the first bearing being disposed in the first bearing chamber and the second bearing being disposed in the second bearing chamber.
12. The electric fan according to claim 11, wherein the motor assembly further comprises a plurality of elastic members; a first group of the plurality of elastic members is arranged between the first bearing and an inner wall of the first bearing chamber and respectively abuts against the first bearing and the first bearing chamber; a second group of the plurality of elastic members is arranged between the second bearing and an inner wall of the second bearing chamber and respectively abut against the second bearing and the second bearing chamber.
13. The electric fan according to claim 3, wherein the electric fan further comprises a deflector, and a first end surface is formed on an end of the bracket adjacent to the diffuser assembly; the deflector is arranged on the first end surface; in the extension direction of the axis of the output shaft, a positive projection of the deflector at least partially overlaps with the positive projection of

the air outlet.

14. The electric fan according to claim 13, wherein the deflector comprises a deflection portion; the deflection portion protrudes toward the air outlet; in the extension direction of the axis of the output shaft, a positive projection of a point on the deflection portion furthest from the first end surface is disposed within the positive projection of the air outlet; wherein a surface of the deflection portion close to the air outlet is curved.

15. The electric fan according to claim 2, wherein the electric fan further comprises a mesh structure; the mesh structure covers a gap between the diffuser assembly and the stator-rotor component, and the mesh structure is connected to the diffuser assembly and the bracket, respectively.

16. The electric fan according to claim 1, wherein the impeller assembly comprises an impeller assembly housing, a first impeller, a reflux component, and a second impeller; the first impeller, the reflux component, and the second impeller are disposed in the impeller assembly housing; the first impeller, the reflux component, and the second impeller are sequentially sleeved on the output shaft; a first flow channel is included in the first impeller, a reflux channel is formed between the reflux component and an inner wall of the impeller assembly housing, and a second flow channel is included in the second impeller; the first flow channel, the reflux channel, and the second flow channel in a sequential communication with each other.

17. The electric fan according to claim 16, wherein the impeller assembly further comprises an elastic sealing ring, and the elastic sealing ring comprises a sealing ring body and an elastic sealing tooth; each of the sealing ring body and the elastic sealing tooth has a ring-shaped structure, and both the sealing ring body and the elastic sealing tooth are sleeved on the output shaft; the sealing ring body is disposed between the reflux component and the elastic sealing tooth and is fixedly connected to the reflux component and the elastic sealing tooth, respectively; the elastic sealing tooth is rotatably connected to the output shaft; in an extension direction of an axis of the output shaft, a width of the elastic sealing tooth is positively correlated with a distance from the elastic sealing tooth to the axis of the output shaft.

18. The electric fan according to claim 16, wherein the impeller assembly further comprises a sealing cotton; at least one of the following is satisfied: the sealing cotton is filled between the impeller assembly housing and the first impeller; and the sealing cotton is filled between the impeller assembly housing and the second impeller.

19. The electric fan according to claim 1, wherein the diffuser assembly comprises a first axial flow diffuser and a second axial flow diffuser; the first axial flow diffuser and the second axial flow diffuser are arranged along an extension direction of an axis of the output shaft; the first axial flow diffuser is sealed and connected to the second axial flow diffuser, and the first axial flow diffuser is in communication with the second axial flow diffuser; an end of the first axial flow diffuser away from the second axial flow diffuser is sealed and connected to the impeller assembly, and the first axial flow diffuser is in communication with the impeller assembly.

20. A terminal device, comprising an electric fan; wherein the electric fan comprises a motor assembly, an impeller assembly, and a diffuser assembly; wherein the motor assembly comprises an output shaft, a bracket, and a stator-rotor component; the bracket and the stator-rotor component are both sleeved on the output shaft; the bracket has an accommodation region, and the stator-rotor component is disposed in the accommodation region; the impeller assembly is disposed on a side of the stator-rotor component and is sleeved on the output shaft; the diffuser assembly is sleeved on the output shaft and is disposed between the stator-rotor component and the impeller assembly; wherein a part of gas passing through the diffuser assembly is configured to flow toward an outside of the bracket, and another part of the gas is configured to flow toward an interior of the stator-rotor component.
